UNIT-4

Current Environmental Issues of Importance; *Global Warming, Green House Effects, Climate Change, Acid Rain, Ozone Layer Formation and Depletion, Population Growth and Automobile pollution, Burning of paddy straw.*

CLIMATE CHANGE

The periodic modification of Earth's climate brought about due to the changes in the atmosphere as well as the interactions between the atmosphere and various other geological, chemical, biological and geographical factors within the Earth's system is called Climate change. Climate is the average weather of an area. It is the general weather conditions, seasonal variations and extremes of weather in region. Such conditions which average over a long period at least 30 years is called climate. However, during the past10000 years of the current interglacial period, the mean average temperature has increased by 0.8 Celsius over 100 year perio

Factors Affecting Climate Change

Natural Factors – affect the climate over a period of thousands to millions of years.Such as –

1. **Continental Drift** – have formed millions of years ago when the landmass began to drift apart due to plate displacement. This impacts climate change due to the change in the landmass's physical features and position and the change in water bodies' position like the change in the follow of ocean currents and winds.
2. **Volcanism** – Volcanic eruption emits gasses and dust particles that last for a longer period causing a partial block of the Sun rays thus leading to cooling of weathers and influencing weather patterns.
3. **Changes in Earth's Orbit –** A slight change in the Earth's orbit has an impact on the sunlight's seasonal distribution reaching earth's surface across the world. There are three types of orbital variations – variations in Earth's eccentricity, variations in the tilt angle of the Earth's axis of rotation and precession of Earth's axis. These together can cause Milankovitch cycles, which have a huge impact on climate and are well-known for their connection to the glacial and interglacial periods.

Anthropogenic Factors – is mainly a human-caused increase in global surface temperature. Such as –

1. **Greenhouse Gasses** – these absorb heat radiation from the sun resulting in an increase in Global Temperature. GHGs mostly do not absorb solar radiation but absorb most of the infrared emitted by the Earth's surface. Read more on Greenhouse Gasses on the given link. Global warming begins with the greenhouse effect, which is caused by the interaction between incoming radiation from the sun and the atmosphere of Earth.
2. **Atmospheric Aerosols –** these can scatter and absorb solar and infrared radiation. Solar radiation scatters and cools the planet whereas aerosols on absorbing solar radiation increase the temperature of the air instead of allowing the sunlight to be absorbed by the Earth's surface. Aerosols have a direct affect on climate change on absorption and reflection of solar radiation. Indirectly it can affect by modifying clouds formation and properties. It can even

be transported thousands of kilometres away through winds and circulations in the atmosphere.

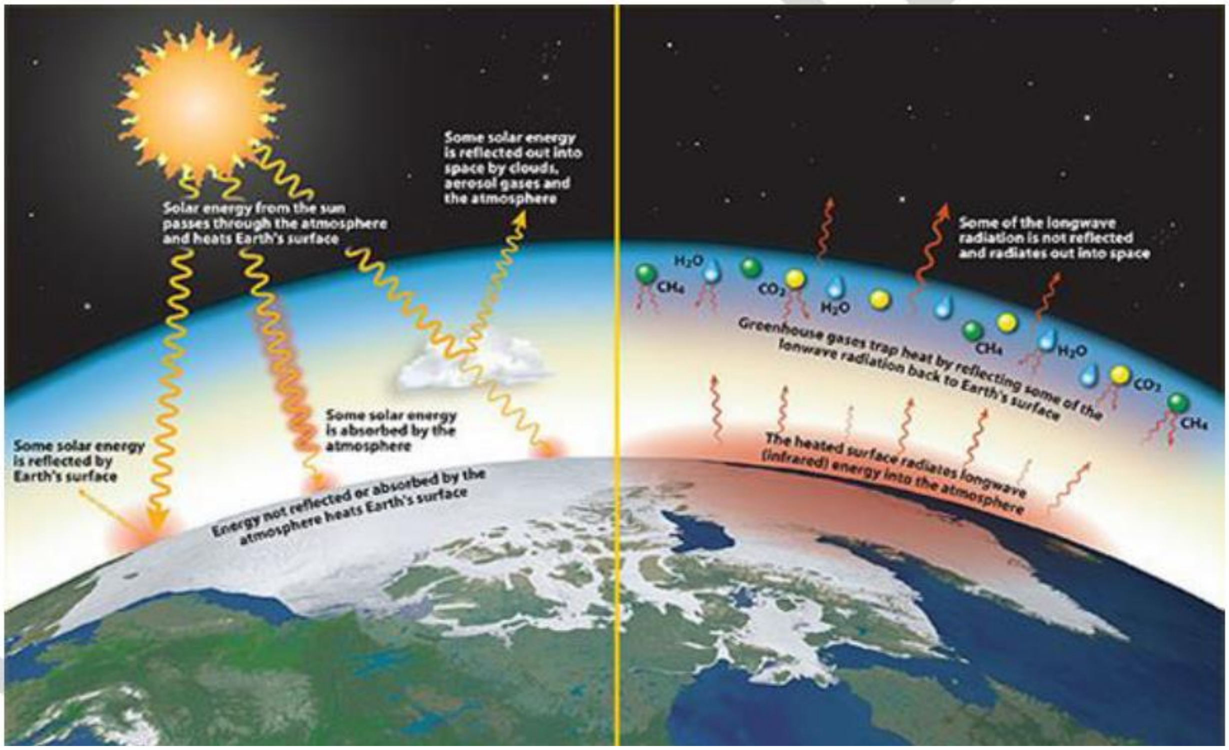
- Read more on Structure of Atmosphere and
- Composition of Atmosphere on the links provided.

3. **Shift in land-use pattern –** Most of the forests and land covers are replaced by agricultural cropping, land grazing, or for Industrial or commercial usage. The clearing of forest cover increases solar energy absorption and the amount of moisture evaporated into the atmosphere.
    - The lower the albedo (reflectivity of an object in space), the more of the Sun's radiation gets absorbed by the planet and the temperatures will rise. If the albedo is higher and the Earth is more reflective, more of the radiation is returned to space, leading to the cooling of the planet.

GREENHOUSE EFFECT AND GLOBAL WARMING

The term 'Greenhouse effect' has been derived from a phenomenon that occurs in a greenhouse. In a greenhouse the glass panel lets the light in, but does not allow heat to escape. Therefore, the greenhouse warms up, very much like inside a car that has been parked in the sun for a few hours.

The greenhouse effect is a naturally occurring phenomenon that is responsible for heating of Earth's surface and atmosphere. Sunlight warms the surface of the Earth. Since the earth cannot store this heat forever, the warm Earth sends energy back into space. The sunlight which hits the Earth's surface is made up of **high energy ultra-violet and visible radiation.** The energy emitted from the surface of the Earth is **infra-red or 'long wave radiation' and is less energetic than sunlight**.



Particles and gases in the air absorb infrared heat radiation. The gases are called **greenhouse gases**. They let the sunlight in, but they don't let the heat radiation from Earth back out into space. They trap the heat near the ground. The greenhouse effect is very important for life on Earth. The average temperature of the Earth is 15 degree Celsius and if there were no greenhouse gases in the air, the average temperature of the Earth would be about 30 degree Celsius lower.

We need a **natural greenhouse effect**. But by putting more and more greenhouse gases into the air, humans have **enhanced** the natural greenhouse effect and are making the Earth warmer. It's not the natural greenhouse effect which is causing global warming, it's the additional greenhouse effect caused by humans which is causing the main culprit.

Greenhouse Gases and their Contribution

Greenhouse gases (GHGs) warm the Earth by absorbing energy and slowing the rate at which the energy escapes to space; they act like a blanket insulating the Earth. Different GHGs can have different effects on the Earth's

warming. Two key ways in which these gases differ from each other are their ability to absorb energy (their "radiative efficiency"), and how long they stay in the atmosphere (also known as their "lifetime"). The most important greenhouse gas is **water vapour** (which accounts for about 60% of the greenhouse effect) but the concentrations of water vapour in the atmosphere have changed much over the past few centuries. So it is unlikely that water vapour is responsible for the observed warming of our planet.

However, human activity has dramatically increased the concentration of **carbon dioxide** in the atmosphere. Carbon dioxide is the most important greenhouse gas in the atmosphere, contributing about 60% of the greenhouse effect (if water vapour is not counted). **Methane** is the second most important greenhouse gas in the atmosphere, contributing about 20% of the greenhouse effect. Concentrations of methane and ozone, which are also strong greenhouse gases, have also increased dramatically since the industrial revolution.

The **Global Warming Potential (GWP)** was developed to allow comparisons of the global warming impacts of different gases. Specifically, it is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO_2). The larger the GWP, the more that a given gas warms the Earth compared to CO_2 over that time period. The time period usually used for GWPs is 100 years. GWPs provide a common unit of measure, which allows analysts to add up emissions estimates of different gases (e.g., to compile a national GHG inventory), and allows policymakers to compare emissions reduction opportunities across sectors and gases. CO_2, by definition, has a GWP of 1 regardless of the time period used, because it is the gas being used as the reference.

Methane (CH_4) is estimated to have a GWP of 28–36 over 100 years. CH_4 emitted today lasts about a decade on average, which is much less time than CO_2. But CH_4 also absorbs much more energy than CO_2. The net effect of the shorter lifetime and higher energy absorption is reflected in the GWP. The CH_4 GWP also accounts for some indirect effects, such as the fact that CH_4 is a precursor to ozone, and ozone is itself a GHG.

Potential Effects of climate change in India

- **Extreme Heat**: India is already experiencing a warming climate. Unusual and unprecedented spells of hot weather are expected to occur far more frequently and cover much larger areas. Under 4°C warming, the west coast and southern India are projected to shift to new, high-temperature climatic regimes with significant impacts on agriculture.
- **Changing Rainfall Patterns**: A decline in monsoon rainfall since the 1950s has already been observed. A 2°C rise in the world's average temperatures will make India's summer monsoon highly

unpredictable. At 4°C warming, an extremely wet monsoon that currently has a chance of occurring only once in 100 years is projected to occur every 10 years by the end of the century. Dry years are expected to be drier and wet years wetter.

- **Droughts**: Evidence indicates that parts of South Asia have become drier since the 1970s with an increase in the number of droughts. Droughts have major consequences. In 1987 and 2002-2003, droughts affected more than half of India's crop area and led to a huge fall in crop production. Droughts are expected to be more frequent in some areas, especially in north-western India, Jharkhand, Orissa, and Chhattisgarh. Crop yields are expected to fall significantly because of extreme heat by the 2040s.
- **Groundwater:** Even without climate change, 15% of India's groundwater resources are overexploited. Falling water tables can be expected to reduce further on account of increasing demand for water from a growing population, more affluent lifestyles, as well as from the services sector and industry.
- **Glacier Melt**: Most Himalayan glaciers have been retreating over the past century. At 2.5°C warming, melting glaciers and the loss of snow cover over the Himalayas are expected to threaten the stability and reliability of northern India's primarily glacier-fed rivers. Alterations in the flows of the Indus, Ganges, and Brahmaputra rivers could significantly impact irrigation, affecting the amount of food that can be produced in their basins as well as the livelihoods of millions of people
- **Sea level rise:** With India close to the equator, the sub-continent would see much higher rises in sea levels than higher latitudes. Sea-level rise and storm surges would lead to saltwater intrusion in the coastal areas, impacting agriculture, degrading groundwater quality, contaminating drinking water, and possibly causing a rise in diarrhoea cases and cholera outbreaks, as the cholera bacterium survives longer in saline water. Kolkata and Mumbai, both densely populated cities, are particularly vulnerable to the impacts of sea-level rise, tropical cyclones, and riverine flooding.
- Apart from this food and energy security are also major concerns. Water scarcity, health hazards among the masses, and migration and political conflicts are expected to grow.

OZONE DEPLETION

A higher than normal concentration of Ozone molecules, called the Ozone layer, is found in Stratosphere. It acts as a shield absorbing ultraviolet radiation from the sun. UV rays are highly injurious to living organisms since DNA and proteins of living organisms preferentially absorb UV rays, and its high energy breaks the chemical bonds within these molecules.

The thickness of the ozone in a column of air from the ground to the top of the atmosphere is measured in terms of **Dobson units** (DU). 1 DU is equivalent to a layer of pure ozone molecules 0.01mm thick.

Ozone gas is continuously formed and destroyed by the action of UV rays on molecular oxygen, resulting in a dynamic equilibrium. Briefly, the process is:

1. Oxygen molecules photodissociate after intaking an ultraviolet photon whose wavelength is shorter than 240 nm. This converts a single O2 into two atomic oxygen radicals.

2. The atomic oxygen radicals then combine with separate O2 molecules to create two O3 molecules.

3. These ozone molecules absorb UV light between 310 and 200 nm, following which ozone splits into a molecule of O2 and an oxygen atom.

4. The oxygen atom then joins up with an oxygen molecule to regenerate ozone.

This is a continuing process that terminates when an oxygen atom "recombines" with an ozone molecule to make two O2 molecules : 2 O3 → 3 O2

However, due to addition of **chlorofluorocarbons** (CFCs) in the atmosphere because of Human activity, this equilibrium has been disturbed.

CFCs **were** widely used as **refrigerants**. CFCs discharged in the lower part of atmosphere move upward and reach stratosphere. In stratosphere, their life-cycle is depicted below:

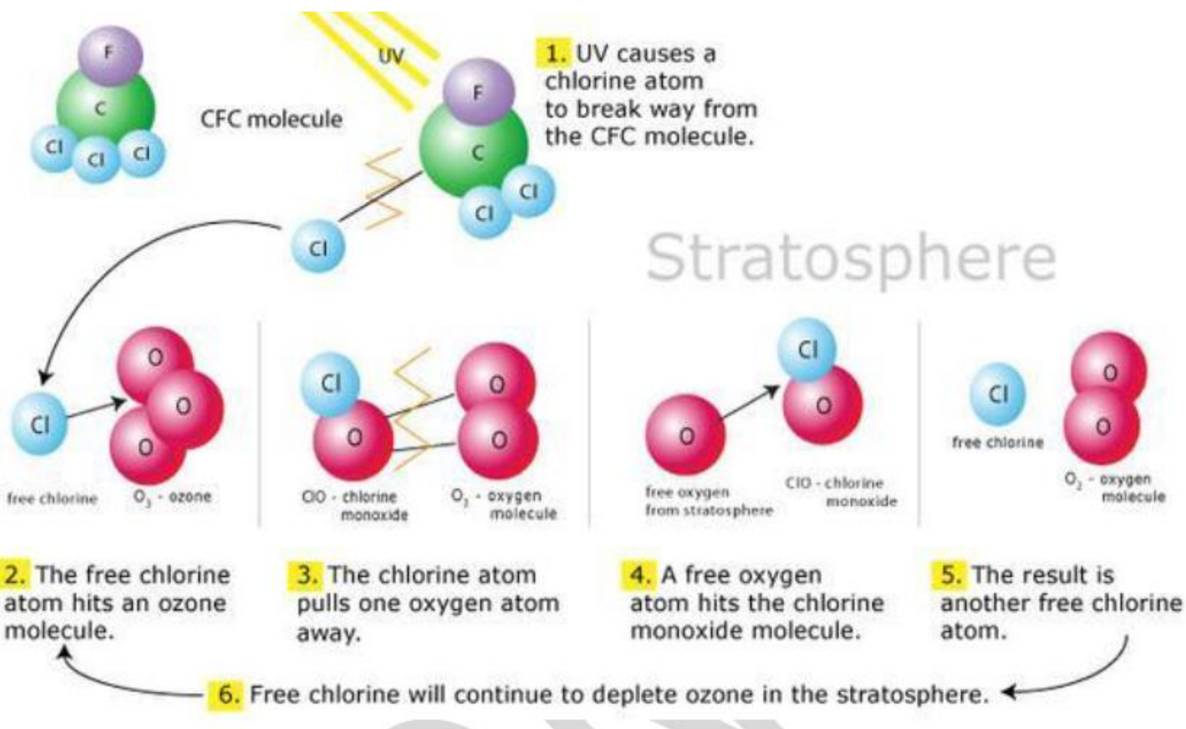


Figure: Catalytic destruction of ozone by atomic halogens

Although ozone depletion is occurring widely in the stratosphere, the depletion is particularly marked over the Antarctic region. This has resulted in formation of a large area of thinned ozone layer, commonly called as the **ozone hole**.

CONSEQUENCES OF OZONE DEPLETION

Humans

Research confirms that high levels of UV rays cause non-melanoma skin cancer. Additionally, it plays a major role in malignant melanoma development. UV is also linked to cataract.

Tropospheric Ozone

Increased surface UV leads to increased tropospheric ozone. Ground-level ozone is generally recognized to be a health risk, as ozone is toxic due to its strong oxidant properties. The risks are particularly high for young children, the elderly, and those with asthma or other respiratory difficulties. Ozone at ground level is produced mainly by the action of UV radiation on combustion gases from vehicle exhausts.

Plants

Plant growths as well as its physiological and developmental process are all affected negatively.

Biogeochemical Cycle

The power of higher UV level affects the natural balance of the gases including the greenhouse gases in the biosphere. Changes in UV level can cause biosphere atmospheric feedback resulting from the atmospheric build-up of these gases.

Acid Rain

Clean rain is slightly acidic naturally but **when the pH of rain falls below 5.6**, we call it acid rain. Emissions of the two air pollutants, nitrogen oxides (NOx) and sulphur dioxide (SO2) are the main reasons for acid rain formation. Nitrogen oxides (NOx = NO + NO2) and sulphur dioxide (SO2) are emitted during fossil fuel combustion and then undergo reactions with water in the air to form the nitric acid (HNO3) and the sulphuric acid (H2SO4) found in acid rain.

Acid rain affects all elements of the environment, surface- and ground-water, soils and vegetation. It negatively affects food chains, reduces biodiversity and damages our world as discussed below:

- When soil becomes acidified, essential nutrients such as calcium (Ca) and magnesium (Mg) are leached out before the trees and plants can use them to grow. This reduces the soil's fertility. In addition, acidification may release1 aluminium from the soil. At high concentrations, aluminium is toxic and damages plant roots. This reduces the plants ability to take up nutrients such as phosphorus, eventually leading to death.
- It leads to acidification of water bodies. Some 14,000 Swedish lakes, located in acidic crystalline rocks, have been affected by acidification with widespread damage to plant and animal life as a consequence.
- Acid precipitation does not usually kill trees directly. Acid deposition destroys the surfaces of the leaves of trees and plants. This damage causes uncontrolled water loss and slows photosynthesis. It reduces the rate at which leaf litter decomposes, causes the death of useful microorganisms present in tree roots and reduces the rate at which soil organisms (including bacteria) respire.
- Soil acidification releases metals that can harm microorganisms in the soil as well as birds and mammals higher up in the food chain. The most sensitive groups include fish, lichens, mosses, certain fungi and small aquatic organisms. Some organisms may be completely eliminated, reducing biodiversity.
- Acid rain also disturbs the natural cycles of sulphur and nitrogen.

Automobile pollution

Automobiles are also a major cause for atmospheric pollution at least in the metro cities. The badly designed and badly maintained automobiles are a major cause of air pollution. This happens because of poor maintenance of the vehicle, and due to this, the engine does not burn the fuel completely or properly. It can also happen because of improper mixture of air and fuel or when poor quality of fuel is used. As a result of the increasing number of vehicles on the roads, air pollution is shifting to the other cities as well. Proper maintenance of automobiles along with the use of lead-free petrol or diesel can reduce the pollutants they emit. Catalytic converters, having expensive metals, namely platinum-palladium and rhodium as the catalysts, are fitted into automobiles for reducing emission of poisonous gases. As the exhaust passes through the catalytic converter, unburnt hydrocarbons are converted into carbon dioxide and water. Also, carbon monoxide and nitric oxide are changed to carbon dioxide and nitrogen gas, respectively. Motor vehicles equipped with catalytic converter should use unleaded petrol because lead in petrol reduces the effectiveness of the catalyst.

Auto Emissions

When emission from automobiles carry unburnt hydrocarbons, it causes air pollution. Pollution from cars comes from by-products of this combustion process (exhaust) and from evaporation of the fuel itself.

Combustion Process

Petrol and diesel fuels are mixtures of hydrocarbons, compounds which contain hydrogen and carbon atoms. In a 'perfect' engine, oxygen in the air converts all the hydrogen in the fuel to water and all the carbon in the fuel to carbon dioxide. Nitrogen in the air remains unaffected. In reality, the combustion process cannot be'perfect', and therefore, automotive engines emit several type of pollutants.

Emission Standards

Governments and regulatory bodies all over the world discuss with automobile companies and list down requirements that set specific limits to the amount of pollutants that can be released into the environment. Many emission standards focus on regulating pollutants released by automobiles (motor cars) and other powered vehicles, but they can also regulate emissions from industry, power plants, small equipment, such as lawn mowers and diesel generators. Frequent policy alternatives to emission standards are technology standards (which mandate the regulation of emissions of nitrogen oxides [NOx], sulphur oxides, particulate matter [PM] or soot, carbon monoxide [CO], or volatile hydrocarbons).

Emission Norms in India

With the increasing number of vehicles on the roads, the possibility of large scale pollution due to these vehicles has also increased. However, the pollution can be reduced considerably if the vehicles are designed and maintained as per regulations. It was only in 1991 that the first stage emission norms came into force for petrol vehicles, and in 1992, for diesel vehicles. From April 1995, mandatory fitting of catalytic converters in new petrol passenger cars sold in the four metros of Delhi, Kolkata, Mumbai and Chennai, along with a supply of Unleaded Petrol (ULP) was brought into effect. Availability of ULP was further extended to 42 major cities and now it is available throughout the country. The emission reduction achieved from pre1989 levels is over 85 per cent for petrol-driven and 61 per cent for diesel vehicles from 1991 levels. In the year 2000, passenger cars and commercial vehicles started meeting Euro I equivalent India 2000 norms. Euro II equivalent Bharat Stage II norms are inforce from 2001 in the four metros of Delhi, Mumbai, Chennai and Kolkata. India is still behind Euro norms by a few years. These are standards followed by European countries. However, with many vehicles manufactured in India now being exported, a beginning has been made, and emission norms are being aligned with Euro standards and vehicular technology is being accordingly upgraded. Indian vehicle manufactures are also working towards bridging the gap between Euro standards and Indian emission norms. Bharat stage emission standards are emission standards instituted by Government of India to regulate the output of air pollutants from internal combustion engine equipments, including motor vehicles. The standards and the timeline for implementation are set by the Central Pollution Control Board under the Ministry of Environment, Forests and Climate Change. **Nowadays, India has adopted Bharat Stage IV norms in the automobile sector. For metro cities, it has become compulsory to use the standard product.**

Burning of Paddy Straw(STUBBLE BURNING)

Stubble burning is a common farming practice around the world and a major cause of air pollution.

Stubble burning is an old and outdated practice that is prevalent in the rice and wheat belt states of India: Haryana, Punjab, and Uttar Pradesh. It is the leading cause of air pollution in the region, especially in the capital, New Delhi.

State governments are actively working to reduce and ultimately eliminate stubble burning, but there's a long journey ahead. On that note, we bring you this comprehensive guide on stubble burning, its meaning, environmental impact, and alternatives.

What is Stubble, and Why Is it Burnt?

Stubble is the straw residue left after harvesting paddy, wheat, and some other grain crops. It remains on the field and takes a long time to decompose. Stubble makes it difficult to plant seeds for the next crop and disturbs the growing cycle.

Moreover, it is not edible for humans and is a poor source of fodder for animals. Stubble has other uses as well, but they are expensive and most farmers can't afford them. Thus, stubble is a nuisance for farmers, and the only viable option left for them is to burn it.

Does Stubble Burning Have Any Benefits?

Stubble burning does have some benefits for farmers. It helps clear weeds, pests, and rodents in the field. It can also reduce nitrogen buildup. However, the biggest benefit and reason behind stubble burning is economics.

Collecting stubble, transporting it, and selling it is just not worth the time, effort, or money for farmers. They are already struggling with a low-income profession like farming, and adding more expenses to it is the last thing they need. Stubble burning is the quickest way of clearing the field before the next planting cycle.

How does stubble burning impact the environment?

Stubble burning has several harmful effects on the environment. It produces toxic gases that not only make it difficult to breathe for nearby residents but also significantly contribute to global warming.

Stubble burning emits pollutants like carbon monoxide (CO), methane (CH_4), carbon dioxide (CO_2), aromatic hydrocarbons, and volatile organic compounds (VOCs), resulting in smog. Stubble burning is the leading cause of air pollution in many parts of the world, including New Delhi.

Stubble burning also affects soil quality. The added carbon from stubble burning and the heat it produces reduce soil fertility and increase erosion.

There is also the risk of fire spreading out of control during stubble burning. Many farmers have even lost their lives to it

What are the alternatives to Stubble Burning?

Stubble can be put to many other uses instead of being burned. Stubble can be treated to create biofuel or particle boards. It can also be used for cattle feed after a treatment, compost manure, roofing, packaging, and paper.

Fast decomposing bio-enzymes are also prevalent in some countries, which quickly degrade the stubble into manure and improve soil composition. One such low-cost microbial bio-enzyme, Pusa decomposer, has been developed for the Indian agricultural industry as well.